Project Coversheet

Full Name	Mitesh Milind Parab
Project Title (Example – Week1, Week2, Week3, Week 4)	Week 4 - Project: Business Intelligence Dashboard for TechHub Retail

Instructions:

Students must download this cover sheet**,** use it as the first page of their project, and then save the entire document as a PDF before submission**.**

Project Guidelines and Rules

1. Formatting and Submission

- Format: Use a readable font (e.g., Arial/Times New Roman), size 12, 1.5 line spacing.
- Title: Include Week and Title (Example - Week 1: Travel Ease Case Study.)
- File Format: Submit as PDF or Word file
- Page Limit: 4–5 pages, including the title and references.

2. Answer Requirements

- Word Count: Each answer should be within 100–150 words; Maximum 800–1,200 words.
- Clarity: Write concise, structured answers with key points.
- Tone: Use formal, professional language.

3. Content Rules

- Answer all questions thoroughly, referencing case study concepts.

- Use examples where possible (e.g., risk assessment techniques).
- Break complex answers into bullet points or lists.

4. Plagiarism Policy

- Submit original work; no copy-pasting.
- Cite external material in a consistent format (e.g., APA, MLA).

5. Evaluation Criteria

- Understanding: Clear grasp of business analysis principles.
- Application: Effective use of concepts like cost-benefit analysis and Agile/Waterfall.
- Clarity: Logical, well-structured responses.
- Creativity: Innovative problem-solving and examples.
- Completeness: Answer all questions within the word limit.

6. Deadlines and Late Submissions

- Deadline: Submit on time; trainees who fail to submit the project will miss the “Certificate of Excellence”

7. Additional Resources

- Refer to lecture notes and recommended readings.
- Contact the instructor or peers for clarifications before the deadline.

YOU CAN START YOUR PROJECT FROM HERE

1. ***Introduction***

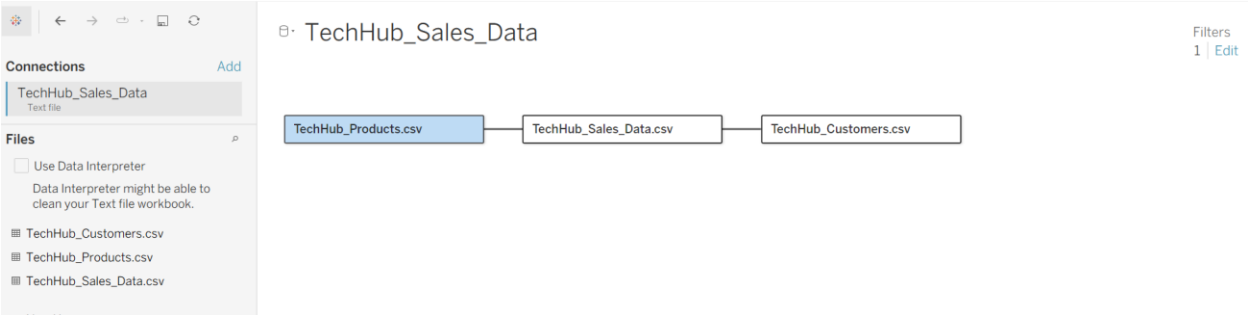
- TechHub Retail is an e-commerce retailer selling technology products through multiple suppliers. The objective of this analysis is to evaluate sales performance, customer behavior, and supplier profitability.
- The purpose of the dashboard is to help management identify profitable products, understand customer demographics, and detect seasonal sales trends to support strategic decisions for 2025 planning.

2. ***Multi Dataset Integration Summary***

- Three datasets (**Sales, Products, Customers**) were connected in Tableau using **Product_ID and Customer_ID relationships**.

- **Calculated Fields Created**
  - **Profit Margin** = Profit / Sales → Identify high-margin products
  - **Average Order Value** = Sales / Orders → Understand customer spending
  - **Inventory Gap** = Stock Quantity – Reorder Level → Detect restocking needs

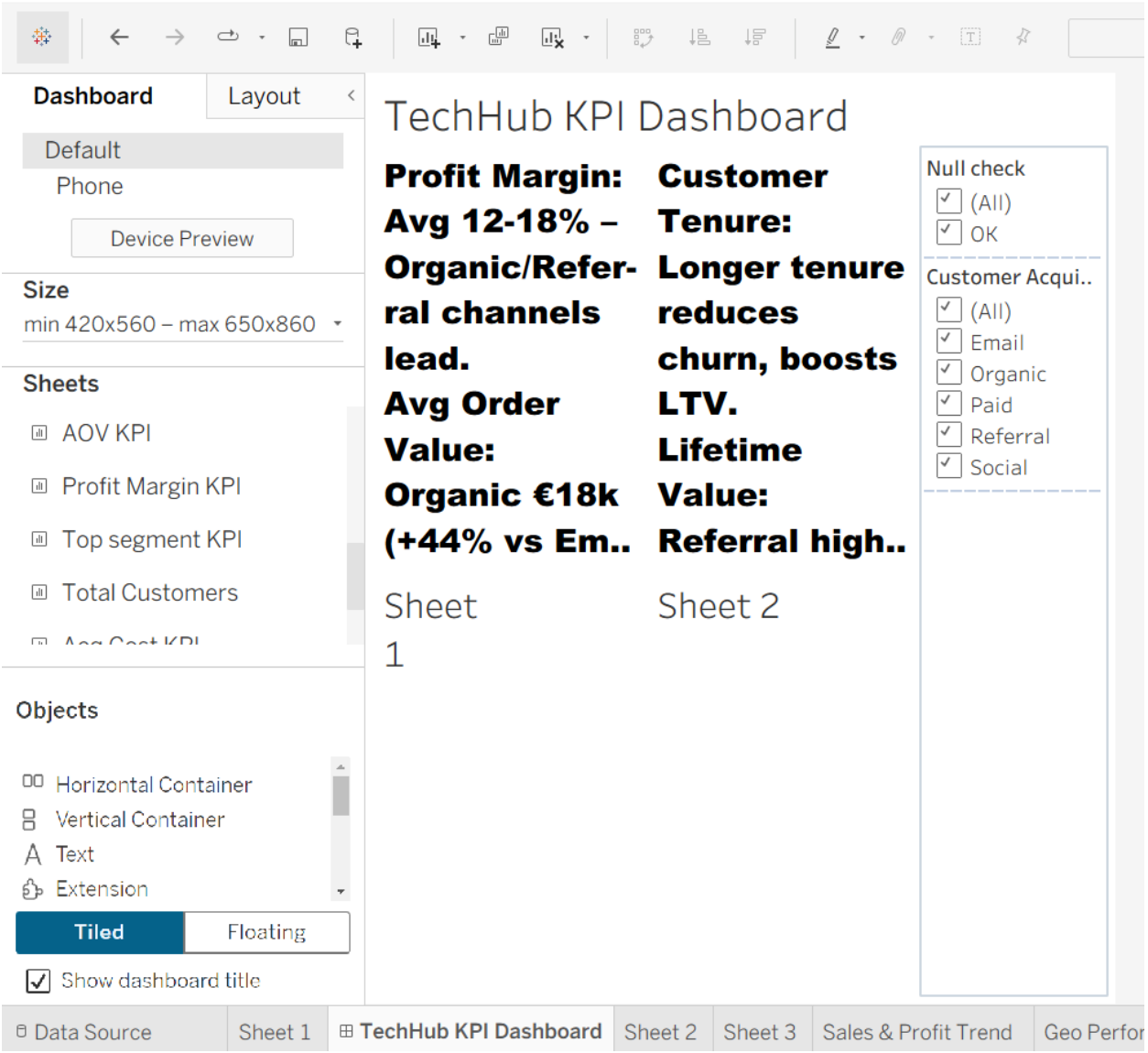
- **Challenges**
  Some tables had **duplicate keys and inconsistent supplier IDs**. These were resolved by **cleaning joins and using relationships instead of inner joins**.

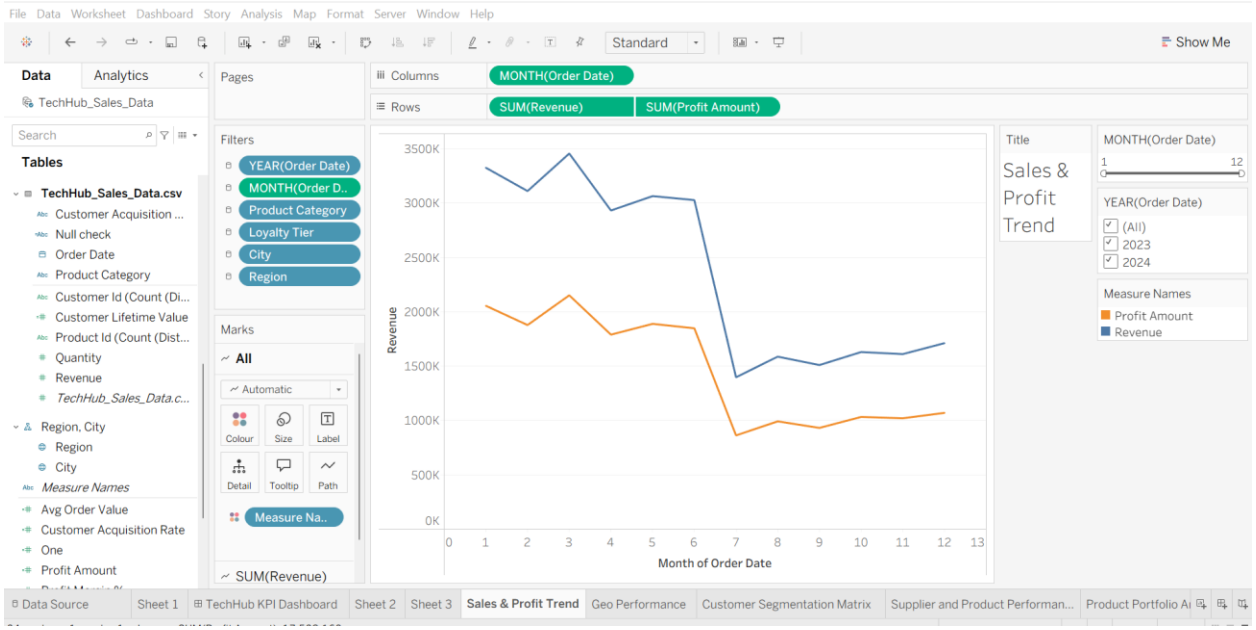


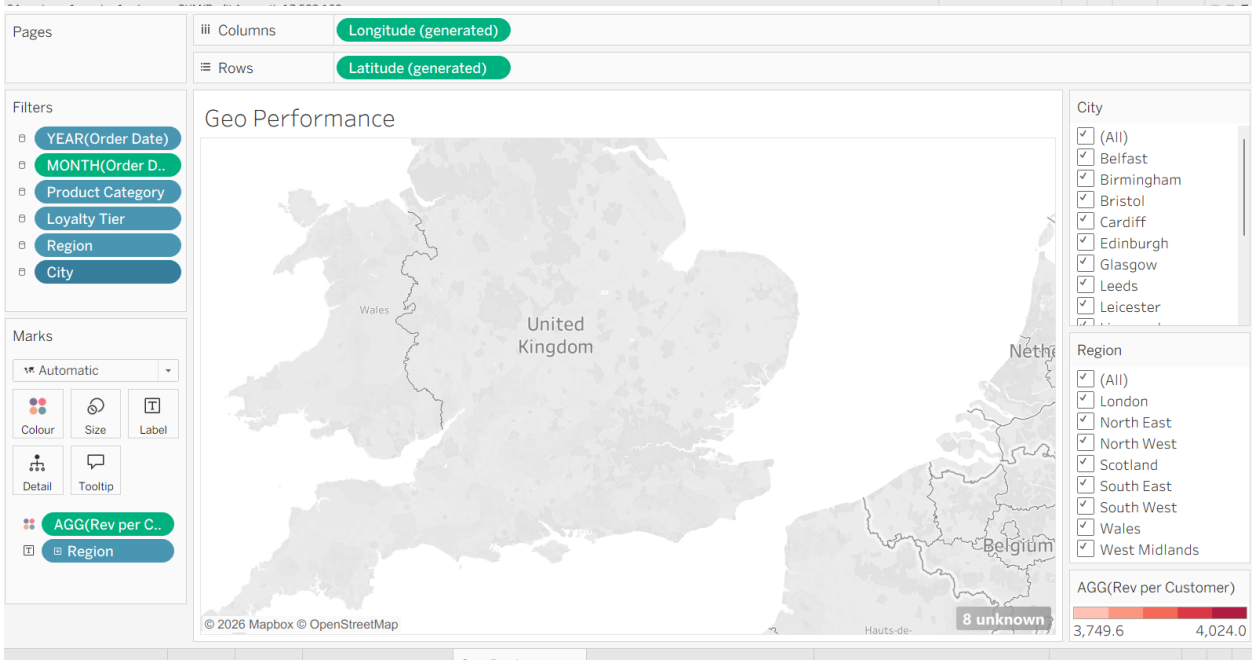
3. ***Dashboard Design Summary***

- The dashboard includes:
  - Sales by Category (bar chart)
  - Supplier Profitability (bar chart)
  - Customer Demographics Analysis (age/location chart)
  - Seasonal Sales Trends (line chart)

- Interactive filters allow users to filter by category, supplier, and time period. Tooltips and highlight actions help executives quickly explore patterns.



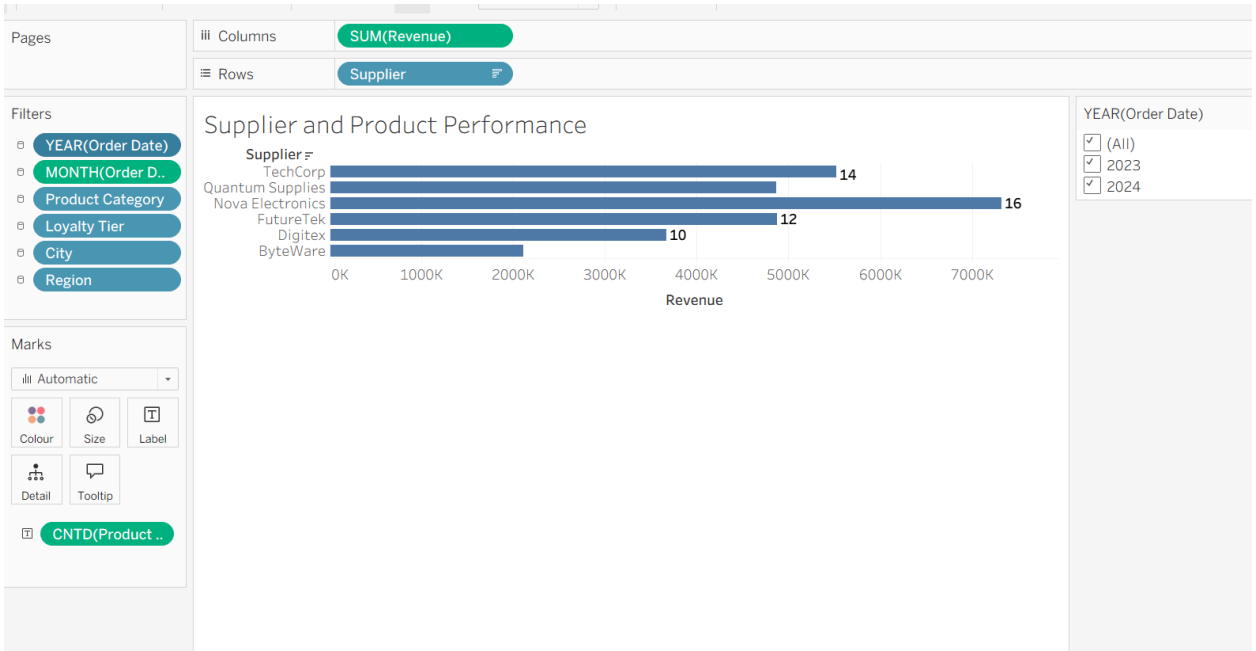




4. ***Key Insights and Finding***

- The dashboard analysis revealed several important patterns in TechHub Retail's sales performance, customer behavior, and supplier contribution.

- First, the **Electronics category generated the highest revenue and profit margins** compared to other product categories. This indicates that electronics products are one of the most valuable segments for the business and play a key role in overall profitability.
- Second, the analysis of customer behavior showed that **premium or loyal customers tend to spend more per purchase**. Their average order value is higher than other customer groups, suggesting that retaining loyal customers can significantly increase revenue.
- Another important finding was related to **seasonal sales patterns**. Sales activity increased noticeably during the **fourth quarter (October–December)**. This trend is likely linked to holiday shopping and promotional events, making this period particularly important for marketing campaigns and inventory planning.
- Finally, the supplier performance analysis revealed that **some suppliers generated high sales but relatively low profit margins**. This suggests that while their products sell well, pricing or supplier costs may be reducing overall profitability. Reviewing supplier agreements or pricing strategies could help improve margins in these cases.



5. ***Business Questions***

1. Which product categories and suppliers offer the best profit margins for 2025 focus?

   The analysis shows that the **Electronics category provides the highest revenue and profit margins**. Some suppliers consistently contribute to higher profitability compared to others. This suggests that TechHub should prioritize partnerships with high-performing suppliers and focus more on profitable product categories when planning its 2025 strategy.

2. How do customer demographics (age, location, loyalty tier) impact purchasing behavior?

   Customer analysis indicates that **loyal or long-term customers tend to spend more per order**, resulting in a higher average order value. Certain customer groups also show more frequent purchases, suggesting that targeted marketing campaigns for these segments could increase overall sales.

3. What seasonal patterns exist across different product categories and regions?

   Sales trends reveal that **sales increase significantly during the fourth quarter (Q4)**. This pattern suggests that holiday periods and promotional seasons strongly influence purchasing behavior. Planning promotions, inventory, and marketing campaigns before this period could help maximize revenue.

4. Which customer acquisition channels deliver the highest lifetime value customers?

   The dashboard indicates that **organic and referral channels tend to generate customers with higher lifetime value** compared to some paid channels. These customers often stay longer and make larger purchases.

5. How does product age (time since launch) correlate with sales performance?

   Sales performance varies depending on product age, with some newer products performing strongly while certain older products continue to maintain stable sales.

6. What are the top 3 strategic recommendations based on integrated data analysis?

   1. Focus on high-performing product categories

   The analysis shows that some categories, such as electronics, generate higher revenue and profit margins. TechHub should focus more on promoting and expanding these high-performing product categories to increase overall profitability.

   2. Strengthen relationships with high-performing suppliers

   Certain suppliers contribute significantly to sales and profit. Building stronger

partnerships with these suppliers and prioritizing their products could help improve sales performance and ensure consistent product availability.

3. Invest more in high-value customer acquisition channels
Customers acquired through organic search and referral channels tend to have higher lifetime value. TechHub should invest more in these channels, such as improving search visibility and referral programs, to attract more valuable customers.

6. Strategic Recommendations

1. Focus on high-performing product categories
The analysis shows that electronics generate the highest revenue and profit margins. TechHub should increase marketing and product availability in these categories.
Timeline: 3–6 months
Impact: Higher sales and improved profitability.

2. Strengthen partnerships with top suppliers
Some suppliers contribute significantly to overall sales. Prioritizing these suppliers and negotiating better agreements can improve supply reliability and profit margins.
Timeline: 6–12 months
Impact: Better supplier performance and stable inventory.

3. Invest in high-value customer channels
Customers from organic and referral channels show higher lifetime value. Increasing investment in these channels can attract more valuable customers.
Timeline: 3–9 months
Impact: Higher customer retention and increased average order value.

4. Prepare for seasonal demand
Sales increase during certain periods of the year. Planning inventory and

promotions before peak seasons can help maximize sales.

Timeline: Before peak seasons

Impact: Higher sales and fewer stock shortages.

7. Critical Reflection

The dashboard effectively supports executive decision-making by presenting key insights on **sales, customers, and supplier performance** in a clear and visual format. Interactive filters allow users to quickly explore trends and identify important business patterns.

However, future BI reports could be improved by adding **more detailed customer segmentation, predictive analysis, and additional historical data** to provide deeper insights and more accurate business forecasting.

8. Data Issues or Risks

Some limitations exist in the dataset used for this analysis. There may be **data imbalance between product categories or suppliers**, which could affect the accuracy of comparisons. In addition, some fields may contain **missing or inconsistent values**, which can influence the reliability of the results.

Another potential risk is **limited historical data**, which may reduce the accuracy of identifying long-term trends and seasonal patterns. These factors should be considered when interpreting the insights from the dashboard.

This project analyzed TechHub Retail's sales, customer behavior, and supplier performance by integrating multiple datasets and presenting the results through an interactive Tableau dashboard. The dashboard helped identify important patterns such as **high-performing product categories, valuable customer segments, seasonal sales trends, and supplier contributions to revenue and profit**.

The analysis showed that **electronics products generate strong revenue and profit margins**, loyal customers tend to spend more per purchase, and sales increase during key seasonal periods. It also highlighted that while some suppliers drive high sales volumes, their profit margins may require further evaluation.

Overall, the dashboard provides a **clear and user-friendly way for decision-makers to explore business performance and identify growth opportunities**. The insights and strategic recommendations from this analysis can support TechHub Retail in improving **marketing strategies, supplier management, and product planning for future business growth**.